

Lemma. Let $\mathcal{Q}: F \rightarrow F'$ be a field Isomorphism, and let $\mathcal{Q}^*: F[x] \rightarrow F'[x]: \sum r_i x^i \mapsto \sum \mathcal{Q}(r_i) x^i$.
 Let $p(x) \in F[x]$ be irreducible, and let $p^*(x) = \mathcal{Q}^*(p(x))$.
 If β is a root of $p(x)$ and β' is a root of $p^*(x)$,
 then, there exists a unique Isomorphism $\overline{\mathcal{Q}}: F(\beta) \rightarrow F(\beta')$ s.t. $\overline{\mathcal{Q}}|_F = \mathcal{Q}$, $\overline{\mathcal{Q}}(\beta) = \beta'$.
 Proof. Since $\mathcal{Q}^*$ is also Isomorphism,

$$\Sigma: F[x]/(p(x)) \rightarrow F[x]/(p^*(x)): f(x) + (p(x)) \mapsto \mathcal{Q}^*(f(x)) + (p^*(x))$$

It maps $c + (p(x)) \mapsto \mathcal{Q}(c) + (p^*(x))$ for all $c \in F$ and $x + (p(x)) \mapsto x + (p^*(x))$, and Isomorphism.

Now, $\overline{\mathcal{Q}}: F(\beta) \xrightarrow{\phi^{-1}} F[x]/(p(x)) \xrightarrow{\Sigma} F[x]/(p^*(x)) \xrightarrow{\phi'} F(\beta')$

$$: \alpha \mapsto (\phi' \circ \Sigma \circ \phi^{-1})(\alpha)$$

'is a field isomorphism leaving F fixed, and β maps to β' .

Lemma. Let $f(x) \in F[x]$ and $D_x f(x) \in F[x]$ be a derivative of $f(x)$. Then,

$f(x)$ has a multiple root α 'if and only if' $D_x f(x)$ has α as a root.

In particular,

$f(x)$ is separable 'if and only if' $(f(x), D_x f(x)) = 1$.

Proof. Suppose that $(x-\alpha)^2$ divides $f(x)$ in the splitting field K of f . Then,

$$f(x) = (x-\alpha)^2 \cdot h(x) \quad \text{for some } h(x) \in K[x].$$

By the property of derivative,

$$D_x f(x) = 2 \cdot (x-\alpha) \cdot h(x) + (x-\alpha)^2 \cdot D_x h(x) = (x-\alpha)(2h(x) + (x-\alpha) \cdot D_x h(x)).$$

Thus proved. Conversely, suppose that both f and $D_x f$ has α as a root. Then,

$$f(x) = (x-\alpha) \cdot h(x) \quad \text{for some } h(x) \in K[x], \text{ and } D_x f(x) = h(x) + (x-\alpha) \cdot D_x h(x).$$

And, by the assumption, $(x-\alpha)$ divides $D_x f$, so divides $h(x)$. Now,

$$f(x) = (x-\alpha)h(x) = (x-\alpha)^2 r(x), \quad \text{where } r(x) = h(x)/(x-\alpha).$$

Now, the last assertion 'is obtained by',

f is separable 'if and only if' f has no multiple root 'if and only if' $\deg(f(x), D_x f(x)) = 0$ 'if and only if' $(f(x), D_x f(x)) = 1$.

Thm. A field of characteristic 0 is perfect.

Proof. Let $f(x) \in F[x]$ be an irreducible polynomial. If $D_x f(x)$ is not zero, then $\deg D_x f < \deg f$.

Therefore, $(f, D_x f) = 1$, because if $(f, D_x f) \neq 1$, then it is contradiction to f is irreducible. So it is separable.

Lemma. A field of characteristic p is perfect iff every element of F has a p th root in F .

Proof. Let F be a field of characteristic p .

Suppose that F is perfect. Let $\alpha \in F$ be a non-zero, non-1. put $f(x) = x^p - \alpha$.

If $f(x)$ has a root in F , then there is nothing to prove.

Assume f has no root in F . Observe that in splitting field K of $f(x)$, put $\beta \in K$ s.t.

$$f(\beta) = \beta^p - \alpha = 0.$$

And, $f(x) = x^p - \alpha = x^p - \beta^p = (x - \beta)^p$ in K . But, if f is reducible in F , then for some $1 \leq k < p$, $f(x) = (x - \beta)^{p-k} \cdot (x - \beta)^k$ and $(x - \beta)^{p-k}, (x - \beta)^k \in F[x]$.

That is, $(-\beta)^k \in F$. Clearly, $(k, p) = 1$, thus $ku + pv = 1$ for some $u, v \in \mathbb{Z}$.

Now, $\beta = \beta^{ku+pv} = (\beta^k)^u \cdot (\beta^p)^v = (\beta^k)^u \cdot \alpha^v \in F$. It is contradiction. But, if f is irreducible then $D_x f = p \cdot x^{p-1} = 0$ implies contradiction with F is perfect. Now, α has a p th root in F .

Conversely, let $f(x) \in F[x]$ be an irreducible. Assume it is inseparable.

Denote $f(x) = a_n x^n + \dots + a_1 x + a_0$. Since $f(x)$ is irreducible and inseparable,

$D_x f(x) = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \dots + a_1$ must be zero, that is,

$$a_k \neq 0 \Leftrightarrow k \in (p)$$

Therefore, we can write

$$f(x) = b_m x^{mp} + b_{m-1} x^{(m-1)p} + \dots + b_1 x^p + b_0$$

Now, by the assumption, each b_i has p th root c_i , thus

$$f(x) = c_m^p \cdot (x^m)^p + c_{m-1}^p \cdot (x^{m-1})^p + \dots + c_0^p$$

$$= (c_m \cdot x^m + c_{m-1} \cdot x^{m-1} + \dots + c_0)^p. \text{ But it is reducible. Contradiction.}$$

Corollary. Every finite field is perfect. (char is p on finite thus Auto.)

Thm. Every finite separable extension is simple extension.

Proof. If F is finite, then the finite extension E/F is also finite. Then, E^X is cyclic, therefore $E = F(\alpha)$ where α is generator of E^X .

Now, assume that F is infinite. Since E/F is finite,

$$E = F(\alpha_1, \dots, \alpha_r) \text{ where } \alpha_1, \dots, \alpha_r \in E.$$

Now, enough to show $F(\beta, \gamma) = F(\alpha)$ for some $\alpha \in E$, and after is solved by the induction.

Let $\beta_1, \dots, \beta_n$ be distinct zeros of $\text{irr}(\beta, F)$, and

$\gamma_1, \dots, \gamma_m$ be distinct zeros of $\text{irr}(\gamma, F)$, in $\bar{F}$. ($\beta_1 = \beta$, $\gamma_1 = \gamma$).

Since E/F is separable, each zeros are multiplicity of 1.

Take $a \in F$ s.t. $a \neq \frac{\beta_i - \beta}{\gamma - \gamma_j}$ for all $1 \leq i \leq n$, $2 \leq j \leq m$, since F is infinite.

Put $\alpha = \beta + a \cdot \gamma$. Then, for all $(i, j) \in \{1, \dots, n\} \times \{2, \dots, m\}$,

$$\alpha = \beta + a \gamma \neq \beta_i + a \gamma_j, \text{ or } \alpha - a \gamma_j \neq \beta_i.$$

Observe that: $\text{irr}(\beta, F)(\alpha - a\gamma) \in F(\alpha)[x]$, and

$$\text{irr}(\beta, F)(\alpha - a\gamma) = \text{irr}(\beta, F)(\beta) = 0.$$

But, $\text{irr}(\beta, F)(\alpha - a\gamma_j) \neq 0$ for all $2 \leq j \leq m$,

That is, $\text{irr}(\beta, F)(\alpha - a\gamma)$ and $\text{irr}(\beta, F)(x)$ have a common factor $(x - \gamma)$ in $F(\alpha)$, thus $\gamma \in F(\alpha)$, so $\beta = \alpha - a\gamma \in F(\alpha)$.